Debriefings in Misinformation Research Should Include Fact-Checks to Correct Misperceptions and Increase Perceived Learning

Journal:	*Political Science Research and Methods*
Manuscript ID	PSRM-2025-0282.R1
Manuscript Type:	Research Note
Date Submitted by the Author:	14-Jan-2026

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Keywords:	Experimental Research, Mass Media and Political Communication

Abstract:	Debriefings are a cornerstone of ethical research, especially when deception is used. In (experimental) misinformation research, this often involves exposing participants to false claims without their awareness. Debriefings both disclose deception and act as interventions, warning participants about misinformation. Yet, unlike such interventions (e.g., fact-checking, media literacy), they reveal that researchers deliberately introduced falsehoods, raising concerns about unintended effects on trust and skepticism. In a pre-registered U.S.-based experiment (N = 896), we compare the effects of no debriefing, a general debriefing, and a specific debriefing that includes fact-checks and source links. We find that only the specific debriefing significantly reduced belief in the false claims that participants were exposed to before the debriefing, with no effect on novel false claims presented after the debriefing. The general debriefing reduced belief in true news, suggesting that explicit warnings may heighten generalized skepticism. Both debriefings improved participants' attitudes toward the study, increasing perceived transparency and reduced feelings of manipulation. These findings suggest that while general warnings may not reduce misperceptions and come with unintended side effects, specific debriefings effectively correct belief in the specific claims they address and foster trust — supporting their use as an ethical and effective practice in misinformation research.

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Abstract

Debriefings are a cornerstone of ethical research, especially when deception is used. In (experimental) misinformation research, this often involves exposing participants to false claims without their awareness. Debriefings both disclose deception and act as interventions, warning participants about misinformation. Yet, unlike such interventions (e.g., fact-checking, media literacy tools) they reveal that researchers deliberately introduced the falsehoods, raising concerns about unintended effects on trust and skepticism. In a pre-registered U.S.-based experiment (N = 896), we compare the effects of no debriefing, a general debriefing, and a specific debriefing that includes fact-checks and source links. We find that only the specific debriefing significantly reduced belief in the false claims that participants were exposed to before the debriefing, with no effect on novel false claims presented only after the debriefing. The general debriefing slightly reduced belief in true news, suggesting that explicit warnings may heighten generalized skepticism. Both debriefings improved participants' attitudes toward the study, increasing perceived transparency and reduced feelings of manipulation. These findings suggest that while general warnings may not reduce misperceptions and may come with small unintended side effects, specific debriefings effectively correct belief in the specific claims they address and foster trust — supporting their use as an ethical and effective practice in misinformation research.

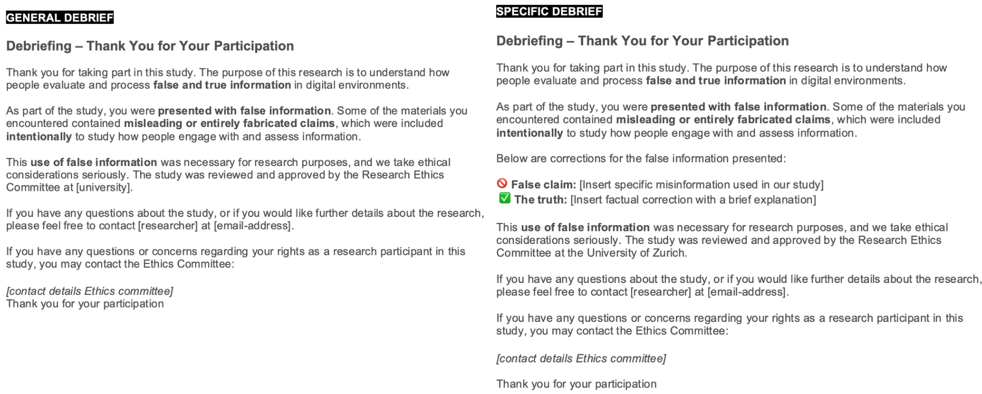
Keywords: Debriefings, Ethics, Misinformation, Trust, Skepticism, Experiments, Deception

1. Introduction

Experimental research frequently employs deception, such as withholding information from participants or exposing them to false claims without revealing that the claims are false. This ensures the internal validity of the experiment and the successful delivery of the treatment (Holmes, 1976; Walster et al., 1967). One prevalent form of deception in misinformation research, central to this study, involves presenting misinformation to participants without their awareness (Clayton et al., 2024; Greene et al., 2023). Given ethical obligations to correct participants' induced false beliefs, effective debriefing becomes essential, not only to correct misperceptions but also to preserve participants' trust and willingness to engage in future research (Oczak and Niedźwieńska, 2007).

However, recent research suggests that interventions correcting misinformation, though effective in reducing false beliefs, can inadvertently increase skepticism toward accurate information and erode trust in credible sources (Hoes et al., 2024; Altay et al., 2024; Thorson, 2024; van der Meer et al., 2023; van Erkel et al., 2024). For instance, Hoes et al. (2024) show that common misinformation interventions (e.g., fact-checking, media-literacy messaging) often prime individuals to approach all information skeptically, diminishing trust in accurate information and that warnings about misinformation can decrease trust in scientists (Hoes et al., 2025). Debriefings in experimental research may produce similar effects: not only do they explicitly warn participants about the presence of misinformation, but they also reveal that participants were deliberately deceived. This combination could amplify feelings of being misled, increasing skepticism toward the study or science in general. At the same time, as Altay et al. (2024) and van Erkel et al. (2024) note, transparency-focused interventions can promote trust when they clearly signal reliability and distinguish false from true content. Debriefings, particularly when specific and detailed, may therefore also enhance perceptions of researcher transparency and foster greater trust and engagement.

Given these nuanced findings, this pre-registered study (see also Appendix A) systematically investigates both potential positive and negative consequences of specific and general debriefing strategies following exposure to misinformation within the study. In line with recent research on debriefings Clayton et al. (2024), we operationalize the difference between a specific and general debriefing such that in the former the false claims used in the study are actively debunked and accompanied by links to fact-checking websites, whereas in the latter they are not. To ensure our treatments reflect actual research practices, we designed them based on debriefing materials extracted from studies Greene et al. (2023) comprehensive review of ethical practices in misinformation research. Greene et al. (2023) documented that both specific and general debriefing approaches remain in active use, though researchers providing detailed debriefings were more likely to report them in publications than those using general approaches, making the true prevalence of general debriefings uncertain. Our treatments were thus designed to reflect real-world variation in debriefing practices currently employed in the field.



GENERAL DEBRIEF

Debriefing – Thank You for Your Participation

Thank you for taking part in this study. The purpose of this research is to understand how people evaluate and process **false and true information** in digital environments.

As part of the study, you were **presented with false information**. Some of the materials you encountered contained **misleading or entirely fabricated claims**, which were included **intentionally** to study how people engage with and assess information.

This **use of false information** was necessary for research purposes, and we take ethical considerations seriously. The study was reviewed and approved by the Research Ethics Committee at [university].

If you have any questions about the study, or if you would like further details about the research, please feel free to contact [researcher] at [email-address].

If you have any questions or concerns regarding your rights as a research participant in this study, you may contact the Ethics Committee:

[contact details Ethics committee]
Thank you for your participation

SPECIFIC DEBRIEF

Debriefing – Thank You for Your Participation

Thank you for taking part in this study. The purpose of this research is to understand how people evaluate and process **false and true information** in digital environments.

As part of the study, you were **presented with false information**. Some of the materials you encountered contained **misleading or entirely fabricated claims**, which were included **intentionally** to study how people engage with and assess information.

Below are corrections for the false information presented:

🚫 **False claim:** [Insert specific misinformation used in our study]
✅ **The truth:** [Insert factual correction with a brief explanation]

This **use of false information** was necessary for research purposes, and we take ethical considerations seriously. The study was reviewed and approved by the Research Ethics Committee at the University of Zurich.

If you have any questions about the study, or if you would like further details about the research, please feel free to contact [researcher] at [email-address].

If you have any questions or concerns regarding your rights as a research participant in this study, you may contact the Ethics Committee:

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Thank you for your participation

Figure 1: Debriefings used as the Treatment Materials. The General Debriefing is depicted on the left. The Specific Debriefing on the right.

Based on previous evidence (Clayton et al., 2024), we hypothesized that specific debriefings will correct false beliefs more effectively than general debriefings. Specifically, we examine belief accuracy for false news that were presented both before and after treatment, as well as for novel false claims shown only post-treatment. The key difference here is that the repeated news items were corrected in the specific debriefing, whereas the novel news items were not. Throughout the paper, we refer to these as 'repeated' false (and true) news items — those that participants rated both before and after the debriefing, and which were explicitly debunked in the specific debriefing, and 'novel' false (and true) news items — those presented only after the debriefing and not mentioned in either debriefing condition. This terminology allows us to distinguish between correction of previously exposed items and potential spillover effects to new (mis)information.

In addition, we explore a series of open-ended research questions regarding broader outcomes that remain largely unexamined in prior work on debriefings (see also Appendix A. While earlier studies have primarily focused on belief correction as the main outcome (Clayton et al., 2024; Greene et al., 2023; Ross et al., 1975; Walster et al., 1967), few have systematically investigated how debriefings affect trust in information, perceptions of transparency, or willingness to participate in future research. Though some have raised theoretical concerns about such consequences Oczak and Niedźwieńska (2007), empirical evidence remains limited. We therefore assess whether debriefings unintentionally heighten skepticism toward accurate information (using both repeated and novel news items), or conversely, enhance trust and attitudes toward the study. We summarize the pre-registered directional hypotheses and open research questions in Appendix B. By simultaneously examining potential harms (e.g., generalized skepticism) and benefits (e.g., trust and transparency), our study offers novel insights into how debriefings can better balance methodological rigor, ethical responsibility, and participant welfare in experimental misinformation research.

2. Methods

The pre-registrations, materials, R scripts, and the data to reproduce the findings are publicly available on OSF. The full pre-registration is also available in Appendix A.

This study received IRB approval. On June 13, 2025, we recruited 900 U.S. participants via the platform Prolific. After excluding four participants who failed the attention check, our final sample consisted of 896 respondents (median age = 37; 482 women and 404 men; median education = college degree). Before being randomly assigned to one of the two treatment groups (specific or general debriefing) or the control group (no debriefing), participants completed a set of demographic questions and a pre-treatment survey in which they rated several true and false claims and indicated their attitudes toward the study, the researchers, and science more broadly. After the treatment, participants again answered questions — some of which repeated earlier items, and others that were new, as detailed below. For ethical reasons, all participants received a final debriefing at the end of the study, regardless of condition.

After the first debriefing, participants in the treatment conditions were asked, "Were you exposed to false or misleading information in this study?" [Yes, No, Not sure]. Participants in the Control Condition answered the same question at a similar point in the survey (without being exposed to the debriefing). This item served as a manipulation check. In the General Debrief condition, 74% of participants reported having been exposed to false or misleading information in the study, compared to 70% in the Specific Debrief and 48% in the Control. Overall, 169 participants in the treatment conditions failed the manipulation check (i.e., did not respond "Yes").

We selected a total of six false claims and six true claims. Of these six, four were part of the pre- and post-treatment survey (repeated), and the two additional items were presented post-treatment only ('novel' items). The false claims were selected from a larger pool of headlines that were gathered by visiting the prominent fact-checking websites Snopes and PolitiFact. The true claims were selected by visiting prominent legacy media outlets such as the New York Times, the Guardian and the Washington Post. Of all the headlines, we selected six true and six false claims balanced in terms of political leaning (i.e., pro Republican or pro Democrat). We based this approach on current best practices in the field Pennycook et al. (2021). The survey, including the true and false claims, can be found on OSF and in Appendix C.

For the true and false claims, and all the questions below, participants answered on a five-point Likert scale ('Strongly disagree', 'Disagree', 'Neutral', 'Agree', 'Strongly agree').

Attitudes towards the study were measured only post-treatment. We did not measure them pre-treatment because participants could not meaningfully evaluate the study's purpose, transparency, or ethical aspects before receiving the debriefing. Measuring them beforehand could also have primed participants or drawn unnecessary attention to the design of the study. Participants indicated whether they agreed with eight statements about the study. Five statements were positive (e.g., "The way this study was conducted was ethically acceptable") and four were negative ("I felt manipulated by how this study was designed"). After reverse coding the negative items, the scale has high internal validity (Cronbach's alpha = 0.78).

Trust in scientists was measured before and after the treatment. Participants indicated whether they agreed with six statements about scientists, three were positive (e.g., "I trust scientists to act responsibly and in good faith") and three were negative ("I am generally skeptical about the intentions of scientists"). After reverse coding the negative items, the scale has high internal validity (Cronbach's alpha = 0.85 pre-treatment and 0.86 post-treatment).

Trust in institutions was measured pre- and post-treatment. Participants indicated whether they agreed with four statements about news organizations, fact-checkers, and the government, such as "I trust news organizations to provide accurate and balanced information". The scale has low internal validity (Cronbach's alpha of 0.59 pre-treatment and 0.66 post-treatment). On OSF, we break down the scale and show that the treatments did not affect any of the specific items of the scale.

As with attitudes toward the study, willingness to participate in future studies was measured only post-treatment, since participants could not meaningfully assess their willingness to participate in similar studies before knowing the study's purpose and procedure. Participants indicated whether they agreed with the following statements: "I am willing to participate in future research studies" and "I would consider taking part in a similar research study that includes exposure to false or misleading information" (Cronbach's alpha = 0.67). On OSF, we break down the scale and show that the treatments did not affect any of the specific items of the scale.

Statistical Analyses

In all the models, we control for age, gender, education, and political orientation. We report the unstandardized beta (b) in the text, while in the Figures we report the standardized beta to facilitate comparison across variables with different scales. We report all the regression tables on OSF. Appendix B summarizes all tested hypotheses, research questions, corresponding comparisons between experimental groups and their results.

To estimate the effect of the debriefings on the perceived accuracy of true and false news, we analyse the data at the response level and conduct linear mixed-effect models with random effects for news item and participant. For news items measured pre- and post-treatment, we interact condition with a term for pre/post, whereas for the novel news items, only measured post-treatment, we do not.

For all the other outcomes, we use ordinary least squares (OLS) linear regressions. For outcomes measured pre- and post-treatment, we estimate differences post-treatment while controlling for pre-treatment levels, whereas for outcomes measured only post-treatment, we do not.

3. Results

In Figure 2, we plot the effect of the debriefings. The blue circles compare the Specific Debriefing to the Control, while the green triangles compare the General Debriefing to the Control.

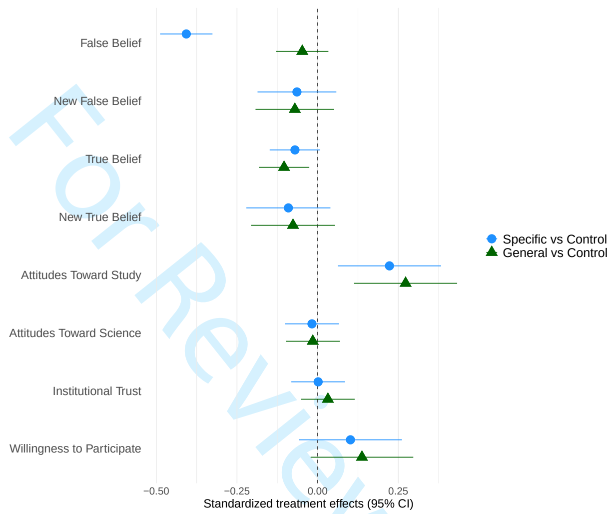

Figure 2: Main effects of the debriefings on all the dependent variables.

Effects of the debriefing on belief in false claims

The debriefings reduced belief in repeated false claims by 0.43 points on the 7-point scale ($p < .001$) compared to the Control Condition – supporting H1a. This effect was driven by the Specific Debriefing, which reduced belief in repeated false claims by 0.78 points ($p < .001$), whereas the General Debriefing did not ($b = 0.09$, $p = .25$). The effect of the Specific Debriefing was 0.69 points larger than that of the General Debriefing ($p < .001$) – supporting H1b. This shows that specific fact-checks reduce belief in false news whereas general warnings about false news do not.

The debriefings did not significantly reduce belief in novel false claims ($b = 0.12$, $p = .21$). As can be seen in Figure 2, the effect sizes are of similar magnitude across the debriefings: 0.13 points for the General Debriefing and 0.12 points for the Specific Debriefing. This null finding indicates that specific debriefings do not produce spillover effects that enhance participants' general ability to identify misinformation beyond the specific false claims that were explicitly corrected.

Effects of the debriefing on true claims

The debriefings reduced belief in repeated true claims by 0.17 points on the 7-point scale ($p = .011$) compared to the Control Condition. In particular, the Specific Debriefing reduced belief in repeated true claims by 0.20 points ($p = .009$) whereas the General Debriefing did not ($b = 0.14$, $p = .08$). However, the effect of the debriefings are not significantly different from each other ($b = 0.07$, $p = .40$). Debriefings may thus prime a general skepticism that affects how participants retrospectively re-evaluate information they encountered during the study, regardless of its veracity. We further reflect on this in the Discussion.

The debriefings did not significantly reduce belief in novel true claims ($b = 0.16$, $p = .15$). As can be seen in Figure 2, the effect sizes are of similar magnitude across the debriefings: 0.15 points for the General Debriefing and 0.18 points for the Specific Debriefing.

Effects of the debriefing on attitudes toward the study

The debriefings improved participants' (positive) attitudes toward the study ($b = 0.14$, $p < .001$) – both the Specific Debriefing ($b = 0.13$, $p = .006$) and the General Debriefing ($b = 0.16$, $p = .001$) did so. In Figure 3, we break down the results by specific attitudes towards the study. Compared to the Control and the General Debriefing, the Specific Debriefing increased participants' feeling that they gained new knowledge by participating in the study ($b = 0.31$, $p < .001$). Compared to the Control, the General Debriefing ($b = 0.21$, $p = .011$) and the Specific Debriefing ($b = 0.24$, $p = .003$) increased the perceived transparency of the study. Compared to the Control, the general debriefing reduced skepticism of researchers' intentions ($b = 0.25$, $p = .002$), feeling of being manipulated ($b = 0.19$, $p = .022$), and anger or frustration associated with the study ($b = 0.23$, $p = .003$) — these effects are not statistically different from the Specific Debriefing.

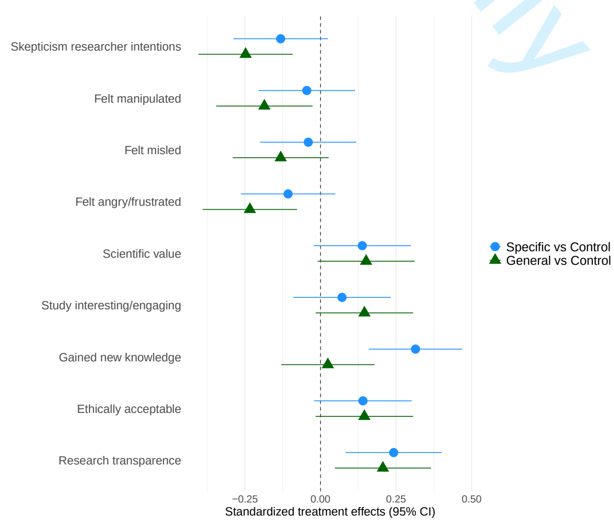

Figure 3: Main effects of the debriefings on attitudes toward the study.

Effects of the debriefing on attitudes toward scientists

The debriefings did not significantly ($b = 0.01$, $p = .66$) influence attitudes towards scientists (e.g., "I trust scientists to act responsibly and in good faith"; both effect sizes are extremely small: $b = 0.01$, $ps > .69$).

Effects of the debriefing on institutional trust

The debriefings did not significantly ($b = - 0.01$, $p = .65$) influence institutional trust (i.e., trust in news organizations, fact-checkers, and the government; both effect sizes are very small: $b < 0.04$, $ps > .47$).

Effects of the debriefing on willingness to participate in surveys

The debriefings did not significantly ($b = - 0.07$, $p = .089$) influence participants' willingness to participate in future surveys ($b < 0.09$, $ps > .08$). On OSF, we break down the scale and show that the treatments did not significantly affect any of the specific items of the scale.

Taken together, we found no evidence that the debriefings affected several outcomes of theoretical concern: attitudes toward science (RQ4), institutional trust (RQ5), or willingness to participate in future studies (RQ6). Similarly, we found no spillover effects to novel items, neither for false news (RQ1) nor true news (RQ2b).

Moderating effect of total number of survey taken on Prolific

The effect of the Specific Debriefing on repeated false claims was stronger among participants who have taken more surveys on Prolific ($b = 0.09$, $p = .032$) – but not for the general debriefing ($b = 0.02$, $p = .57$). Moreover, the positive effects of the debriefings on attitudes towards the study were stronger among participants who have taken more surveys on Prolific ($b = 0.06$, $p = .006$) – for both the Specific Debriefing ($b = 0.05$, $p = .029$) and the General Debriefing ($b = 0.06$, $p = .013$). We found no other statistically significant differences. We expected the effect of the debriefing to be weaker, not stronger, among participants who have more experience with surveys.

Heterogenous treatment effects

We found no statistically significant heterogenous treatment effect by true or false claims (repeated and novel), or by the political orientation of the participants. Compared to the Control, the Specific Debriefing was more effective at correcting belief in repeated false claims among less educated ($b = 0.27$, $p < 0.001$) and older participants ($b = 0.12$, $p = 0.031$). The Specific Debriefing also increased positive attitudes towards the study slightly more among women than men compared to the Control ($b = 0.20$, $p = 0.033$). The general debriefing increased positive attitudes towards the study more among less educated participants compared to the Control ($b = 0,13$ $p = 0.004$).

4. Conclusion & Discussion

This study offers a comprehensive assessment of how different debriefing strategies affect participants in experimental misinformation research. While prior work has mainly focused on belief correction, we extend this literature by examining not only whether debriefings effectively correct misperceptions, but also how they shape participants' attitudes toward the study, trust in science, willingness to participate in future research, as well as their belief in true information. By investigating both intended and unintended consequences—including the possibility that debriefings heighten generalized skepticism—we provide a more complete picture of how debriefings function in the context of misinformation studies. Focusing on the increasingly common practice of exposing participants to misinformation without their awareness, we compare the effects of a general warning and a specific corrective debriefing. Our findings offer both reassuring and cautionary insights for researchers navigating the ethical and methodological trade-offs involved in the use of deception.

Consistent with prior work on fact-checking and debunking interventions (Clayton et al., 2024), specific debriefings improved belief accuracy regarding the exact falsehoods presented during the study. In contrast, general debriefings — though ethically compliant — did not reduce belief in false information. This finding reinforces the growing consensus that vague or broad warnings about misinformation are insufficient to correct misperceptions and that specific, claim-level corrections remain a best practice when addressing false beliefs.

However, our findings also reveal important boundary conditions. The Specific Debriefing showed no spillover effect to novel false claims, indicating that debriefings function as targeted corrections of the specific false beliefs that were induced during the study. This absence of transfer effects suggests that participants do not extract generalizable lessons about identifying misinformation from exposure to corrections, even when those corrections are detailed and accompanied by credible source information. From an ethical standpoint, this is reassuring. Specific debriefings effectively fulfill their primary obligation to correct any false beliefs that may be the results of false claims that researchers deliberately included.

Yet our findings also reveal potential unintended consequences. The general debriefing slightly reduced belief in true news that participants had seen before, suggesting that even ethically motivated transparency can prime generalized skepticism. While this is in line with recent concerns that interventions aimed at debunking misinformation can erode trust in accurate content or sources (Hoes et al., 2024; Thorson, 2024; Altay et al., 2024), our findings refine that concern. Specifically, the similarity in effect size across the general and specific debriefing conditions suggests that this reduction in belief in true news is not primarily driven by the act of debunking itself, but by shared elements of both debriefings — namely, the warning about misinformation and the revelation of intentional deception. Our design cannot isolate which component drives this effect, but the pattern suggests that warning participants about misinformation — especially when framed as having been deliberately introduced by the researchers — may undermine confidence in adjacent or truthful content.

Importantly, however, our findings also point to positive consequences. Both debriefings increased perceived transparency and improved attitudes toward the study. The general debriefing also lowered feelings of manipulation and frustration. These results echo recent work suggesting that interventions can foster trust when they are perceived as transparent and clearly delineate truth from falsehood (Altay et al., 2024; van Erkel et al., 2024). Compared to the general debriefing, the specific debriefing was particularly effective in helping participants feel they had gained new knowledge, further supporting the idea that transparent, claim-level corrections can enhance the perceived value of research participation. Contrary to concerns about backlash from deception, we found no negative effects on trust in scientists, institutions, or willingness to participate in future studies. These findings speak directly to longstanding methodological concerns in experimental economics, where the no-deception rule is maintained primarily to preserve participants' trust and ensure the continued viability of experimental methods (Jamison et al., 2008; Ortmann and Hertwig, 2002). Our results suggest that deception - even when explicitly communicated - need not undermine future experimental validity.

Taken together, these findings offer a more nuanced understanding of how debriefings function — not merely as ethical afterthoughts but as active interventions with meaningful consequences. While the corrective power of specific debriefings is clear, their broader informational and psychological effects deserve more attention. Future work should explore how different framing choices — e.g., apologetic versus assertive tones, attribution of blame, or source emphasis — may shape these effects. Moreover, experimental designs that allow finer isolation of the warning versus correction components are needed to disentangle their respective roles. Additionally, future studies should include behavioral measures — such as actual participation in a second follow-up survey — to assess attrition and willingness to re-engage more robustly than through self-reports alone.

Ultimately, our results suggest that debriefings can both promote accuracy and foster trust, while also introducing risks of overgeneralized skepticism. Ethical experimental design must therefore strike a careful balance between maximizing transparency and learning, while minimizing confusion and doubt.

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Appendix A. Pre-Analysis Plan

Date: 12.06.2025

RESEARCH QUESTIONS

1. What is the effect of general, specific and no debriefings on belief accuracy and updating, trust, skepticism, engagement with and attitudes towards scientific research?

HYPOTHESES AND RESEARCH QUESTIONS

Belief Accuracy

Hypotheses and research questions on the effectiveness of the debriefing

H1a. The debriefings will work as intended and reduce belief in false claims present in the debriefing: Participants in the Debriefing conditions will report lower belief in false claims than participants in the Control.

lmer(score_False ~ condition (baseline = Control) * Pre_Post + age + gender + education + Political_Orientation + (1|NewsItemID) + (1|ParticipantsID))

H1b. The detailed debriefing will be more effective than the general debriefing: Participants in the Specific Debriefing condition will report lower belief in false claims than participants in the General Debrief.

lmer(score_False ~ condition (baseline = *SpecificDebrief*) * Pre_Post + age + gender + education + Political_Orientation + (1|NewsItemID) + (1|ParticipantsID))

RQ1. Will the debriefing also reduce belief in NEW false claims not present in the debriefing?

lmer(score_NewFalse ~ condition (baseline = Control) + age + gender + education + Political_Orientation + (1|NewsItemID) + (1|ParticipantsID))
lmer(score_NewFalse ~ condition (baseline = *SpecificDebrief*) + age + gender + Political_Orientation + education + (1|NewsItemID) + (1|ParticipantsID))

RQ2. Will the debriefing have unintended consequences on true news? For instance, participants in the Debriefing conditions may report lower belief in true claims than participants in the Control.

Testing RQ2 on true news items present in the debriefing:
lmer(score_True ~ condition (baseline = *Control*) * Pre_Post + age + gender + education + Political_Orientation + (1|NewsItemID) + (1|ParticipantsID))
lmer(score_True ~ condition (baseline = *SpecificDebrief*) * Pre_Post + age + gender + education + Political_Orientation + (1|NewsItemID) + (1|ParticipantsID))

Testing RQ2 on NEW true news items:
lmer(score_NewTrue ~ condition (baseline = *Control*) + age + gender + education + Political_Orientation +(1|NewsItemID) + (1|ParticipantsID))
lmer(score_NewTrue ~ condition (baseline = *SpecificDebrief*) + age + gender + Political_Orientation +education + (1|NewsItemID) + (1|ParticipantsID))

Trust (generic science)

***RQ4**: How will the debriefings affect attitudes toward science?*

We will create an index of attitudes toward science composed of positive attitudes (e.g., trust in scientists, belief in ethical conduct, transparency) and negative attitudes (e.g., skepticism about scientists' intentions, belief that research is manipulative or misleading). The negative attitudes will be reverse coded.

To increase power we will merge the General and Specific Debrief conditions – but if they differ statistically we will not merge them.

> lm(Mean_Trust_Positive_Post ~ Mean_Trust_Pre + condition (baseline = *Control*) + Political_Orientation + age + gender + education).

We will also report the results for individual items.

Institutional Trust beyond Science

***RQ5**: Will the debriefings reduce institutional trust (e.g., in news organizations, government, and fact-checkers)?*

To increase power we will merge the General and Specific Debrief conditions – but if they differ statistically we will not merge them.

> lm(Mean_Trust_Post ~ Mean_Trust_Pre + condition (baseline = *Control*) + Political_Orientation + age + gender + education).

Research question on the potential unintended consequences of the debriefing

RQ2. Will the debriefing have unintended consequences on true news? For instance, participants in the Debriefing conditions may report lower belief in true claims than participants in the Control.

Testing RQ2 on true news items present in the debriefing:

> lmer(score_True ~ condition (baseline = *Control*) * Pre_Post + age + gender + education + Political_Orientation + (1|NewsItemID) + (1|ParticipantsID))
> lmer(score_True ~ condition (baseline = *SpecificDebrief*) * Pre_Post + age + gender + education + Political_Orientation + (1|NewsItemID) + (1|ParticipantsID))

Testing RQ2 on NEW true news items:

> lmer(score_NewTrue ~ condition (baseline = *Control*) + age + gender + education + Political_Orientation +(1|NewsItemID) + (1|ParticipantsID))
> lmer(score_NewTrue ~ condition (baseline = *SpecificDebrief*) + age + gender + Political_Orientation +education + (1|NewsItemID) + (1|ParticipantsID))

Research question on effects of the debriefings on attitudes and trust
Trust (study specific)

RQ3: *How will the debriefings affect attitudes toward the study?*

We will create an index of attitudes toward the study composed of positive attitudes (e.g., perceived transparency, ethicality, value of the research) and negative attitudes (e.g., feeling misled, manipulated, frustrated). The negative attitudes will be reverse coded.

To increase power we will merge the General and Specific Debrief conditions – but if they differ statistically we will not merge them.

> lm(Mean_Positive_Attitudes ~ condition (Control *vs* Debriefings) + age + gender + Political_Orientation + education).

We will also report the results for individual items.

Trust (generic science)
RQ4: *How will the debriefings affect attitudes toward science?*

We will create an index of attitudes toward science composed of positive attitudes (e.g., trust in scientists, belief in ethical conduct, transparency) and negative attitudes (e.g., skepticism about scientists' intentions, belief that research is manipulative or misleading). The negative attitudes will be reverse coded.

To increase power we will merge the General and Specific Debrief conditions – but if they differ statistically we will not merge them.

> lm(Mean_Trust_Positive_Post ~ Mean_Trust_Pre + condition (baseline = *Control*) + Political_Orientation + age + gender + education).

We will also report the results for individual items.

Institutional Trust beyond Science
RQ5: *Will the debriefings reduce institutional trust (e.g., in news organizations, government, and fact-checkers)?*

To increase power we will merge the General and Specific Debrief conditions – but if they differ statistically we will not merge them.

> lm(Mean_Trust_Post ~ Mean_Trust_Pre + condition (baseline = *Control*) + Political_Orientation + age + gender + education).

Future Participation
RQ6: *Will the debriefings reduce participants' willingness to participate in future studies?*

To increase power we will merge the General and Specific Debrief conditions – but if they differ statistically we will not merge them.

lm(Willingness_to_Participate_Post ~ condition (baseline = *Control*) + Political_Orientation + age + gender + education).

Research question on effects of the debriefings on HTEs
Heterogeneous Treatment Effects
RQ7*: Will the treatment effects documented above be moderated by participants' experience on Prolific? For instance, the debriefings could have stronger effects on participants who have completed fewer online surveys.*

To increase power we will merge the General and Specific Debrief conditions – but if they differ statistically we will not merge them.

lm(DV ~ condition (baseline = *Control*) * log(N_Prolific_Studies) + Political_Orientation + age + gender + education).

PARTICIPANTS

We will recruit 900 (300 per group) US-based participants via Prolific Academic, balanced by ideology (Democrat/Republican/Independent).

EXCLUSIONS

We will not exclude participants who failed the manipulation check, but we will report the number of participants who failed the manipulation check and also report the results among only participants who passed the manipulation check.

We will exclude participants who fail the attention check.

DESIGN

We have three conditions:

1. General Debrief
2. Specific Debrief
3. No Debrief

Participants are randomly assigned to one of three conditions. For ethical reasons, all participants will receive a final debrief, identical to the specific debrief in this study (see below), at the end of the experiment.

1. **GENERAL DEBRIEF**

Debriefing – Thank You for Your Participation

Thank you for taking part in this study. The purpose of this research is to understand how people evaluate and process **false and true information** in digital environments.

As part of the study, you were **presented with false information**. Some of the materials you encountered contained **misleading or entirely fabricated claims**, which were included **intentionally** to study how people engage with and assess information.

This **use of false information** was necessary for research purposes, and we take ethical considerations seriously. The study was reviewed and approved by the Research Ethics Committee at [university].

If you have any questions about the study, or if you would like further details about the research, please feel free to contact [researcher] at [email-address].

If you have any questions or concerns regarding your rights as a research participant in this study, you may contact the Ethics Committee:

[contact details Ethics committee]
Thank you for your participation

2. SPECIFIC DEBRIEF

Debriefing – Thank You for Your Participation

Thank you for taking part in this study. The purpose of this research is to understand how people evaluate and process **false and true information** in digital environments.

As part of the study, you were **presented with false information**. Some of the materials you encountered contained **misleading or entirely fabricated claims**, which were included **intentionally** to study how people engage with and assess information.

Below are corrections for the false information presented:

🚫 **False claim:** [Insert specific misinformation used in our study]
✅ **The truth:** [Insert factual correction with a brief explanation]

This **use of false information** was necessary for research purposes, and we take ethical considerations seriously. The study was reviewed and approved by the Research Ethics Committee at the University of Zurich.

If you have any questions about the study, or if you would like further details about the research, please feel free to contact [researcher] at [email-address].

If you have any questions or concerns regarding your rights as a research participant in this study, you may contact the Ethics Committee:

[contact details Ethics committee]

Thank you for your participation

Appendix B. Summary of pre-registered hypotheses, questions and comparisons

	Hypotheses and research questions	**Comparison**	**Support and result**
H1a	Lower belief in false claims	Debriefs *vs* control	Supported: the debriefs reduced belief in false claims
H1b	Lower belief in false claims	Specific debriefing *vs* General debriefing	Supported: the specific debriefing was more effective than the general debriefing
RQ1	Effect on belief in new false claims?	Debriefs *vs* control	No significant effect
RQ2a	Effect on belief in true news?	Debriefs *vs* control	The debriefs significantly reduced belief in true news
RQ2b	Effect on belief in new true news?	Debriefs *vs* control	No significant effect
RQ3	Effect on positive attitudes towards the study?	Debriefs *vs* control	The debriefs significantly increased positive attitudes towards the study
RQ4	Effect on attitudes towards science?	Debriefs *vs* control	No significant effect
RQ5	Effect on trust in institutions?	Debriefs *vs* control	No significant effect
RQ6	Effect on willingness to participants in future studies?	Debriefs *vs* control	No significant effect
RQ7	Heterogenous effect by participants' experience on Prolific?	Debriefs *vs* control	No significant effect

Appendix C. Survey

Consent Form

General information

We appreciate your interest in participating in this study. You have been invited to take part as you have registered for this task on Prolific. Please note that you may only participate in this study if you are 18 years of age or over. Please read through the information provided and sign the consent form below, if you wish to proceed.

The aim of this study is to see how you evaluate information. As a participant, you will be asked to read several claims and rate them. Before that, you will be asked some basic questions about your news use and political interest. You will then be presented with a debrief, after which we will again ask your opinion on a number of topics. This should take about 10 minutes. No background knowledge is required.

Do I have to take part?

Please note that your participation is voluntary. You may withdraw at any point during the questionnaire for any reason, before submitting your answers, by pressing the 'Exit' button / closing the browser. However, we only remunerate participants who complete the entire study.

Will I be compensated?

Yes. Participants who successfully complete the study will be paid 2 USD (9 GBP/hour).

How will your data be used?

Your answers will be completely anonymous, and we will take reasonable steps to keep them confidential.

Your data will be stored in a password-protected file and may be used in academic publications. Your IP address will not be stored. Research data will be stored for a minimum of three years after publication or public release.

Who will have access to your data?

[University] is the data controller. Your information may be shared with other researchers internal to the project for data analysis and interpretation purposes. Responsible members of [university] and funders may also be given access to data for monitoring and/or auditing of the study to ensure we are complying with guidelines, or as otherwise required by law.

We would like your permission to use your anonymised data in future studies, and to share data with other researchers (e.g. in online databases). Any personal information that could identify you will be removed or changed before files are shared with other researchers or results are made public.

Who is leading this study?

This study is part of [project]. The principal researcher are [names] who are affiliated with [university]. This project is being completed under the supervision of [name].

This project has been reviewed by, and has received ethics clearance through, [university] Research Ethics Committee.

What if there is a problem?

If you have a concern about any aspect of this project, please reach out to the lead researcher [name, email-address] who will do their best to answer your query. The researcher should acknowledge your concern within 10 working days and give you an indication of how they intend to deal with it. If you remain unhappy or wish to make a formal complaint, please contact the relevant Chair of the Research Ethics Committee at [university]:

[contact details ethics committee]

The Chair will seek to resolve the matter in a reasonably expeditious manner.

Demographics

EDUCAT.
What is the highest level of school you have completed?

1. None, or grades 1-8
2. High school incomplete (grades 9-11)
3. High school graduate (grade 12 or GED certificate)
4. Technical, trade or vocational school AFTER high school
5. Some college, no 4-year degree (includes associate degree)
6. College graduate (B.S., B.A., or other 4-year degree)
7. Post-graduate training/professional school after college (toward a Master's degree or Ph.D., Law or Medical school)

POLITICAL NEWS.
On a typical day, how much time would you say you spend reading the news (either on a mobile phone or a computer?

1. Less than 30 minutes
2. 30-60 minutes
3. 1-2 hours
4. 2-3 hours
5. 3+ hours

POLITICAL INTEREST.
How interested are you in politics?

1. Not at all interested
2. Not very interested

3. Somewhat interested
4. Very interested
5. Extremely interested

EXCLUSIONS.

Attention Check
To check whether you are reading the questions, click on the second answer from the top.

- Never
- Last year
- Last month
- Last week
- Yesterday
- Other

MAIN OUTCOMES

1. Belief Accuracy News Headlines (pre- and post treatment)

Please read each news headline carefully and indicate how accurate you believe it is. Some headlines may be true, others false. Use the scale from 1 (definitely false) to 7 (definitely true) to rate each one based on your best judgment

6 False

- The United States has the sickest population in the world, U.S. Secretary of Health and Human Services Robert Kennedy Jr. told Fox News (pro republican)
- Ford plans to move four factories back to the U.S., along with 25,000 high-paying jobs, all thanks to U.S. President Donald Trump's 2025 tariffs (pro republican)
- “US Debt Clock” numbers prove “DOGE has saved taxpayers $64 billion dollars in just 17 days.” (pro republican)
- Taylor Swift was removed from the 2025 Grammy nominations after a request by Elon Musk. (pro democrat)
- “Elon Musk announced that he will not sell Tesla Cybertruck to anyone belonging to the ‘pride’ community.” (pro democrat)
- Photos prove U.S. President-elect Donald Trump failed to place his right hand over his heart during former President Jimmy Carter's state funeral. (pro democrat)

6 True

- First lady Melania Trump opens White House gardens to all Americans (pro republican)
- RFK Jr. seeks to tighten loophole allowing chemicals in U.S. food supply (pro republican)
- In February 2025, Republican U.S. Rep. Claudia Tenney introduced a bill that would make President Donald Trump's birthday a federal holiday. (pro republican)
- Tech billionaire and adviser to U.S. President Donald Trump, Elon Musk, called Social Security a "Ponzi scheme." (pro democrat)

- White House asks supreme court to allow deportations under wartime law (pro democrat)
- Trump's E.P.A. to Rewrite Rules Aimed at Averting Chemical Disasters (pro democrat)

2. Belief in Controversial Items related to Misinformation, Social Media and Artificial Intelligence (only pre-treatment)

The following statements are often heard in public debates about social media, misinformation, and artificial intelligence.

For each statement, please answer:

1. **Indicate how much you personally agree or disagree** with the statement (1 = completely disagree, 7 = completely agree).

Social Media

- Social media is a major cause of depression and suicide
- Social media users are trapped in echo chambers where they only see views they agree with
- Social media is mainly responsible for the misinformation crisis
- Social media has made people more polarized than ever before.
- The internet has made people more informed than ever before.
- Generally speaking, people encounter more diverse opinions online than offline

Misinformation

- Generally speaking, people encounter more misinformation than true information online
- Generally speaking, false information spreads faster than true information on social media
- False information spreads six times faster than the truth online
- Misinformation has a substantial influence on people's behavior
- Conspiracy theories are more widespread and influential now than ever before
- People can tell the truth from falsehoods
- People tend to be excessively skeptical rather than gullible

Artificial Intelligence

- Because of generative AI tools, soon we won't be able to tell truth from falsehoods
- AI is a major threat to elections worldwide
- Deepfakes will soon make video evidence unreliable.
- AI-generated content will soon dominate online discourse.

3. Trust - Study Specific (post-treatment only)

Positive

- I believe this study contributes to scientific research. (1 = strongly disagree, 5 = strongly agree)
- I found participating in this study interesting and engaging. (1 = strongly disagree, 5 = strongly agree)
- I feel that I gained new knowledge by participating in this study. (1 = strongly disagree, 5 = strongly agree)
- The way this study was conducted was ethically acceptable. (1 = strongly disagree, 5 = strongly agree)
- The researchers were transparent about their goals and methods. (1 = strongly disagree, 5 = strongly agree)

Negative

- This study made me more skeptical about the intentions of researchers. (1 = strongly disagree, 5 = strongly agree)
- I felt manipulated by how this study was designed. (1 = strongly disagree, 5 = strongly agree)
- I felt misled by the use of false information in this study. (1 = strongly disagree, 5 = strongly agree)
- I felt angry or frustrated when I learned that some information in this study was false. (1 = strongly disagree, 5 = strongly agree)

4. Trust - Generic (pre- and post-treatment)

- I trust scientists to act responsibly and in good faith. (1 = strongly disagree, 5 = strongly agree)
- I believe scientific research is generally conducted in an ethically acceptable way. (1 = strongly disagree, 5 = strongly agree)
- Scientists are generally transparent about their goals and methods. (1 = strongly disagree, 5 = strongly agree)
- I am generally skeptical about the intentions of scientists. (1 = strongly disagree, 5 = strongly agree)
- Scientific research often involves manipulative techniques. (1 = strongly disagree, 5 = strongly agree)
- I believe people are often misled by scientific studies. (1 = strongly disagree, 5 = strongly agree)

5. Institutional Trust Beyond Science (pre- and post-treatment)

- I trust news organizations to provide accurate and balanced information. (1 = strongly disagree, 5 = strongly agree)
- I trust government officials to act in the public's best interest. (1 = strongly disagree, 5 = strongly agree)
- I believe independent fact-checkers are trustworthy sources of information. (1 = strongly disagree, 5 = strongly agree)
- It is difficult to know which sources of information to trust. (1 = strongly disagree, 5 = strongly agree)

6. Future Study Participation & Engagement (post-treatment only)

- I am willing to participate in future research studies. (1 = strongly disagree, 5 = strongly agree)
- I would consider taking part in a similar research study that includes exposure to false or misleading information (1 = strongly disagree, 5 = strongly agree)

7. Belief in 'new' True/False News (post-treatment only)

2 True

- Melania Trump joins push for law on deepfakes, nonconsensual porn (pro republican)
- Trump says he 'couldn't care less' about higher car prices (pro democrat)

2 False

- "Malia Obama received $2.2 million in funds from USAID." (pro republican)
- Video shows "leaked audio" of Vice President JD Vance criticizing Elon Musk and saying "he's making us look bad." (pro democrat)

8. Manipulation Check (post-treatment, all groups)

Were you exposed to false or misleading information in this study?
Scale: Yes / No / Not sure

FINAL DEBRIEF (all groups)

Debriefing – Thank You for Your Participation

Thank you for taking part in this study. The purpose of this research is to understand how people evaluate and process **false and true information** in digital environments.

As part of the study, you were **presented with false information**. Some of the materials you encountered contained **misleading or entirely fabricated claims**, which were included **intentionally** to study how people engage with and assess information.

Below are corrections for the false information presented:

🚫 **False claim:** [Insert specific misinformation used in our study]
✅ **The truth:** [Insert factual correction with a brief explanation]

This **use of false information** was necessary for research purposes, and we take ethical considerations seriously. The study was reviewed and approved by [Institutional Review Board].

If you have any questions about the study, or if you would like further details about the research, please feel free to contact [Researcher's Name] at [email address].

Thank you for your participation

Reviewer: 1

This study explores the downstream consequences of different types of misinformation-experiment debriefings on belief and on trust.
The authors compare the effects of a "general" and "specific" debriefing. The general debriefing tells the audience that they have been exposed to false information, but not what that information was. The specific debriefing gives an actual factual correction.
Is this type of general debriefing frequent? I can only bring my own experience to bear, but I have conducted around twelve different "misinformation correction" studies, with five different IRBs (including co-authored pieces), and every one has required us to offer a specific correction to any misinformation that we gave to participants. Thus, I am somewhat surprised to see the "general" debrief proposed as a common approach. That being said, I may very well be wrong about this -- perhaps such "general" debriefings are quite frequent. This is an empirical question, and one that could be answered by systematically coding the debriefing procedures for experiments that expose participants to misinformation, and reaching out to researchers who do not post the debriefings publically.

AUTHORS:

We thank the reviewer for raising this important methodological question. To design our treatments, we extracted and analyzed debriefing materials from studies reported in Greene et al. (2023): https://psycnet.apa.org/record/2023-30594-001. The authors had already classified debriefings as specific or general, with coding materials accessible at: https://osf.io/756nv/files/8c3xj?view_only=712e902800944febb16e389a4ea8eefd.

We based our treatment design on debriefing materials we could retrieve from their systematic review. Importantly, this revealed inconsistencies between reported and actual practices: some studies reported using specific debriefs but did not include the materials; others reported using specific debriefs but the described procedures aligned more closely with general debriefs. We also reached out to Clayton et al. (2024) (the most recent work explicitly comparing specific and general debriefings) to obtain their materials.

Greene et al.'s systematic review demonstrates that while specific debriefings are more common than general ones, researchers providing detailed debriefings were more likely to report them in publications than those using general debriefings. This reporting bias means the true prevalence of general debriefings may be underestimated. Given the systematic approach Greene et al. used in retrieving and coding debriefing materials, we conclude both approaches remain in active use, though the precise distribution is uncertain.

In response to this feedback, we have revised the Introduction (page 2, paragraph 2) by adding three sentences that clarify our treatment development process.

Following the sentence describing how we operationalize the two debriefing types, we now state:

> **"To ensure our treatments reflect actual research practices, we designed them based on debriefing materials systematically extracted from studies in Greene et al.'s (2023) comprehensive review of ethical practices in misinformation research. Greene et al. documented that both specific and general debriefing approaches remain in active use, though researchers providing detailed debriefings were more likely to report them in publications than those using general approaches, making the true prevalence of general debriefings uncertain. Our treatments were thus designed to reflect real-world variation in debriefing practices currently employed in the field."**

My more substantive comment is about the actual findings. As far as I can tell, the only difference between the two types of corrections is in "belief in misinformation" -- which makes sense given that the specific debriefing provides corrections of the misinformation provided previously. To put this another way: participants were asked to rate the accuracy of false headlines. Then, in the "specific debriefing" condition, they were given fact-checks telling them those same headlines were wrong. Then, immediately afterwards, they were asked to evaluate the headlines again, and those in the "specific debriefing" condition were (unsurprisingly) more accurate than those who were not given the fact-checks. In other words, when we embed a fact-check in a debriefing, it makes people more accurate. This is consistent with a great deal of previous research showing that fact-checks are effective at reducing belief in misinformation.

Thus, to summarize, the authors find no support for the following pre-registered hypotheses/RQs:
H1a
RQ1
RQ2a
RQ2b
RQ5
RQ6
RQ7
They do find support for the following hypotheses/RQs:
H1b
RQ3

I want to be very clear that null results are still a contribution to knowledge and the fact that only two out of the nine pre-registered tests is significant should not reduce this article's chances of publication. That being said, the (many) null results receive very little attention. It would be helpful to include a specific table listing all the hypotheses, whether they are being tested with the two corrections pooled versus the control or whether the two types of debriefs were being tested against each other, and then whether each was supported or null.

AUTHORS:

We appreciate the reviewer's thoughtful engagement with our pre-registered design and their emphasis that null results contribute valuable knowledge. We want to clarify an important point about our findings. The reviewer states we found support for "two out of nine pre-registered tests," but we actually found support for four hypotheses/RQs:

- **H1a: Supported - Debriefings (pooled) reduced belief in repeated false claims ($b = 0.43$, $p < .001$)**
- **H1b: Supported - Specific debriefing more effective than general (difference = 0.69 points, $p < .001$)**
- **RQ2a: Significant effect found - Debriefings reduced belief in repeated true news ($b = 0.17$, $p = .011$)**
- **RQ3: Supported - Debriefings improved attitudes toward study ($b = 0.14$, $p < .001$)**

The remaining outcomes (RQ1, RQ2b, RQ4, RQ5, RQ6, RQ7) yielded null findings.

We want to respectfully push back on characterizing these results as "only two (or four) out of nine" tests showing effects. We deliberately pre-registered many outcomes as open-ended research questions rather than directional hypotheses precisely because the literature provided insufficient basis for strong predictions about potential unintended consequences. The purpose of RQs 1, 2b, 4, 5, 6, and 7 was to test whether debriefings produce harmful spillover effects, and our null findings provide reassuring evidence that they largely do not (except for the modest effect on repeated true news in RQ2a, which we discuss).

That said, we agree these null findings deserve explicit attention. The reviewer requested a summary table listing all hypotheses, comparisons, and results. Due to space constraints in the main text, we have included this comprehensive table in the Appendix, and include it below for convenience. We have also added a paragraph in the Results section that explicitly summarizes the pattern of null findings, noting that debriefings showed no spillover to novel items and no negative effects on trust in science, institutions, or willingness to participate in future research.

We believe this approach gives appropriate weight to null results while managing the space limitations of the main manuscript.

	Hypotheses and research questions	**Comparison**	**Support and result**
H1a	Lower belief in	Debriefs *vs* control	Supported: the

	false claims		debriefs reduced belief in false claims
H1b	Lower belief in false claims	Specific debriefing *vs* General debriefing	Supported: the specific debriefing was more effective than the general debriefing
RQ1	Effect on belief in new false claims?	Debriefs *vs* control	No significant effect
RQ2a	Effect on belief in true news?	Debriefs *vs* control	The debriefs significantly reduced belief in true news
RQ2b	Effect on belief in new true news?	Debriefs *vs* control	No significant effect
RQ3	Effect on positive attitudes towards the study?	Debriefs *vs* control	The debriefs significantly increased positive attitudes towards the study
RQ4	Effect on attitudes towards science?	Debriefs *vs* control	No significant effect
RQ5	Effect on trust in institutions?	Debriefs *vs* control	No significant effect
RQ6	Effect on willingness to participants in future studies?	Debriefs *vs* control	No significant effect
RQ7	Heterogenous effect by participants' experience on Prolific?	Debriefs *vs* control	No significant effect

In making this paper, I strongly recommend the authors revisit the pre-registration to ensure that they are reporting all the pre-registered analyses. For example, in several places in the pre-registration they propose pooling the two types of debriefs, but no such analysis appears in the main text. The pre-registration also commits to reporting the percentage of respondents who fail the manipulation check, abd does not do this.

AUTHORS:

We now report all pre-registered analyses, i.e., we added all the pooled analyses that we pre-registered. The previous version of the manuscript only included the pre-registered non-pooled analyses, and the reviewer is correct that we should stick to the pre-registration fully. Below we provide an example of a change, where we first report the pooled effect of the treatments on true news and then report the results by treatment.

> **"The debriefings reduced belief in repeated true claims by 0.17 points on the 7-point scale (p = .011) compared to the Control Condition. In particular, the specific Debriefing reduced belief in repeated true claims by 0.20 points (p = .009) whereas the General Debriefing did not (b = 0.14, p = .08)."**

We apologize for overlooking reporting the percentage of respondents who failed the manipulation check, we now do so:

> **"After the first debriefing, participants in the treatment conditions were asked, "Were you exposed to false or misleading information in this study?" [Yes, No, Not sure]. Participants in the Control Condition answered the same question at a similar point in the survey (without being exposed to the debriefing). This item served as a manipulation check. In the General Debrief condition, 74% of participants reported having been exposed to false or misleading information in the study, compared to 70% in the Specific Debrief and 48% in the Control. Overall, 169 participants in the treatment conditions failed the manipulation check (i.e., did not respond "Yes")."**

I would recommend reframing the paper as a whole to clarify the central finding, which is that debriefs that include fact-checks of previously presented misinformation reduce belief in that misinformation. The description of the findings throughout the paper is somewhat misleading -- specific debriefs do not, as the authors say, "reduce belief in false news" (which implies that it improves accuracy more broadly), but rather, fact-checking specific claims reduces belief *in those claims.* Indeed, the authors cleverly embed a design that lets them test whether the specific debrief "spills over" and increases discernment among novel false claims. They find quite clearly that it does not.

AUTHORS:

We thank the reviewer for this feedback, which has led us to reframe key aspects of the paper. The reviewer is correct that our initial framing conflated correction of specific false beliefs participants had been exposed to with broader improvements in discernment or general reduction in susceptibility to misinformation.

Changes made:

1. **Title: Changed from "Reduce False Beliefs" to "Correct Misperceptions". We believe this better captures that we are correcting specific false beliefs participants hold rather than broadly reducing susceptibility to misinformation.**
2. **Abstract: Now explicitly states effects are on "the false claims that participants were exposed to before the debriefing, with no effect on novel false claims presented only after the debriefing."**
3. **Introduction: we added explicit definition of "repeated" versus "novel" items, establishing this terminology for the rest of the paper**
4. **Results: Consistently uses "repeated false/true claims" and "novel false/true claims" throughout. Added explicit interpretation that null effects for novel items indicate lack of spillover effects that would enhance general discernment.**
5. **Discussion: Added paragraph explicitly discussing the boundary condition (that specific debriefings show no spillover to novel false claims) and interpreting this as indicating debriefings function as targeted corrections rather than interventions that enhance general ability to identify misinformation.**

Smaller notes:
It would be helpful to include the OSF link to the pre-registration rather than requiring reviewers to search for it independently.

AUTHORS: Of course, we have added a separate link for the pre-registration:
https://osf.io/yautp/files/39nu7?view_only=ae733a1b2bc04ce3a0f569532aca46aa
We also included the pre-registration in the Appendix.

Reviewer: 2

This manuscript does an excellent job cataloguing potential risks of misinformation exposure in experimental studies. Even though many designs include factual corrections that often negate misinformation effects with respect to beliefs, the authors rightfully point out that misinformation can have spillover effects on trust in genuine information and scientists more broadly. Following Clayton et al. (2024) and others, the authors examine how specific debriefing approaches that target study-specific claims versus more general debriefing procedures affect a variety of outcomes including beliefs, attitudes, and trust.

AUTHORS:

We thank the reviewer for this positive note!

Overall, the authors find general debriefing messages underperform specific debriefing messages across a variety of dimensions. Consistent with Clayton et al. (2024), the authors find that specific debriefs return beliefs back to levels observed in the control condition, whereas the effects of misinformation on false beliefs persist when general debriefing messages are used. Furthermore, specific debriefs produced more positive attitudes regarding the study and perceptions of research transparency. One problematic finding for specific debriefing messages is that decreases in "true news" are also observed, which may reflect broader warnings about misinformation that were also included in the specific debriefing message.

This is a well-executed and important study that provides additional nuance to previous findings and further reaffirms the importance of offering specific corrections to study-specific misinformation. I also appreciate the future directions that the authors describe. For example, it would be useful to know through a panel design whether specific debriefs encourage more participation in future studies due to the positive downstream consequences they observe for research transparency. Though the self-reported data suggest otherwise, the panels used in this study could be leveraged to examine actual behavior. Indeed, it is worth noting that these boosts in perceptions of research transparency are occurring relative to the control, which suggests that participants especially appreciate the specific debriefing message. From the perspective of how ethics boards should navigate experimental research on misinformation, the study largely suggests that such research may be permissible so long as specific corrections are offered, which allows us to continue learning about an important topic where observational designs may have trouble distinguishing between selection bias and causal effects.

We thank the reviewer for these thoughtful comments and the positive assessment of our study's contribution. We appreciate the recognition of our findings' importance for both understanding debriefing effects and guiding ethical research practices.

The reviewer raises an excellent point about using panel designs to examine actual behavioral outcomes, such as whether specific debriefings encourage more participation in future studies. We agree this would be a valuable extension of our work. However, we were unable to implement such a design in the current study because all participants received a final debriefing at the end of the study, regardless of their experimental condition. This was an ethical requirement as we could not leave any participants without correction of the false beliefs we had induced, even those in the control condition. As a result, we do not have a true control group that could be followed over time to observe differential participation rates, as all participants ultimately received information about which claims were false.

That said, we fully agree this represents an important direction for future research, and we have already incorporated this as a valuable future direction in our discussion. Specifically, we note in the manuscript:

> **"Additionally, future studies should include behavioral measures — such as actual participation in a second follow-up survey — to assess attrition and willingness to re-engage more robustly than through self-reports alone."**

A well-designed panel study that could ethically maintain differential conditions over time (perhaps by varying the *timing* or *mode* of debriefings rather than whether they occur at all) would indeed provide valuable behavioral evidence to complement our self-report measures. We appreciate the reviewer highlighting this opportunity and hope our work provides a foundation for such follow-up investigations.

If I were to modify anything about the paper, I'd say that there is potential heterogeneity by claim and political affiliation that could be unpacked. Readers will likely want to know if there are cases where the specific debriefing messages perform better or worse, and if there are subgroups that are more or less likely to respond positively to the debriefing messages. Further, instead of pointing readers to the OSF page, an appendix should be included that documents how questions were measured, debriefing materials, and the various analyses they describe in the paper.

AUTHORS: Of course, we have added a separate link for the pre-registration:
https://osf.io/yautp/files/39nu7?view_only=ae733a1b2bc04ce3a0f569532aca46aa
We also included the pre-registration in the Appendix.

We have added an Appendix to the paper with the full survey, which includes all questions and their measurements as well as consent and debriefing materials. However, given the high number of regression tables (16 for the main analyses; 41 for all the pre-registered analyses reported in the article and many more if including the heterogenous treatment effects) we believe they are more

conveniently accessed and read in the three HTML documents publicly available on OSF.

As suggested by the reviewers we have investigated potential heterogenous treatment effects and report them in a new section at the end of the results:

> **"We found no statistically significant heterogenous treatment effect by true or false claims, or by the political orientation of the participants. Compared to the Control, the specific debriefing was more effective at correcting belief in false news among less educated (b = 0.27, p $<$ 0.001) and older participants (b = 0.12, p = 0.031). The specific debriefing also increased positive attitudes towards the study slightly more among women than men compared to the Control (b = 0.20, p = 0.033). The general debriefing increased positive attitudes towards the study more among less educated participants compared to the Control (b = 0,13 p = 0.004)."**